Type MS · What a log means: reading values and switching to index form

Answer 1 of 5 · Question 8 part b.i · 1 mark

2021 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

8 (a)	$\log_a n = \log_a 3(2n-1)$	M1
	$n = 3(2n-1)$	M1
	$n = \frac{3}{5}$	A1 (3)
(b)(i)	$x = p^3$	B1 (1)
(b)(ii)	$\log_p y - \log_p 2^3 = 4 \Rightarrow \log_p\left(\frac{y}{2^3}\right) = 4$ or $\log_p\left(\frac{y}{8}\right) = 4$	M1
	$\frac{y}{2^3} = p^4 \Rightarrow \left(y = 2^3 p^4 \text{ or } 8p^4\right)$	M1
	$xy = 8p^7$	M1A1 (4)
	ALT (b)(ii)	
	$\log_p x + \log_p y - 3\log_p 2 = 4 + 3 \Rightarrow \log_p\left(\frac{xy}{2^3}\right) = 7$	{M1}
	$\frac{xy}{2^3} = p^7$	{M1}
	$xy = 8p^7$	{M1A1} (4)
		Total 8 marks

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Answer 1 of 5 · Question 8 part b.i · 1 mark

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Part	Mark	Notes
(a)	M1	Uses the addition law of logs correctly $\log_a n = \log_a 3 + \log_a (2n-1) \Rightarrow \log_a n = \log_a 3(2n-1)$ Accept also $\log_a n = \log_a 3 + \log_a (2n-1) \Rightarrow 0 = \log_a \left(\frac{n}{3(2n-1)}\right) = (\log_a 1)$
	M1	For obtaining a linear equation from their log equation and attempting to find a value for n. $n = 3(2n-1)$ leading to a numerical value for n
	A1	For $n = \frac{3}{5}$
(b)(i)	B1	For $x = p^3$
(b)(ii)	M1	For stating that $3\log_p 2 = \log_p 8$ or $\log_p 2^3$ and for using the addition law correctly to combine the LHS: $\log_p y - \log_p 2^3 = 4 \Rightarrow \log_p \left(\frac{y}{2^3}\right) = 4$ or $\log_p \left(\frac{y}{8}\right) = 4$
	M1	Correctly removes logs on both sides to obtain: $\frac{y}{2^3} = p^4 \Rightarrow \left(y = 2^3 p^4 \text{ or } 8p^4\right)$
	M1	For correctly finding the product of their x and their y: $xy = 'p^3' \times '8p^4'$
	A1	For the correct answer of $xy = 8p^7$
	ALT	
	M1	For stating that $3\log_p 2 = \log_p 8$ or $\log_p 2^3$ and states $\log_p x + \log_p y - 3\log_p 2 = 3 + 4$ Uses the addition law correctly to combine the LHS $\log_p x + \log_p y - 3\log_p 2 = 4 + 3 \Rightarrow \log_p \left(\frac{xy}{2^3}\right) = 7$
	M1	Correctly remove logs on both sides to obtain: $\frac{xy}{2^3} = p^7$
	M1	Correctly rearrange their expression to make xy the subject
	A1	For the correct answer of $xy = 8p^7$

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Answer 1 of 5 · Question 8 part b.i · 1 mark

2021 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · What a log means: reading values and switching to index form

Answer 2 of 5 · Question 3 part a · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

3 (a)	$\log_3 9 = 2$	B1 [1]
(b)	$\log_3 9t = \log_9\left(\frac{12}{t}\right)^2 + 2 \Rightarrow \log_3 9 + \log_3 t = 2\left(\log_9 12 - \log_9 t\right) + 2$ $\log_3 9 + \log_3 t = 2\left(\frac{\log_3 12}{\log_3 9} - \frac{\log_3 t}{\log_3 9}\right) + 2$ $\Rightarrow \log_3 9 + \log_3 t = \log_3 12 - \log_3 t + 2$ $\Rightarrow 2\log_3 t = \log_3 12 \Rightarrow \log_3 t^2 = \log_3 12$ $\Rightarrow t^2 = 12 \Rightarrow t = 2\sqrt{3}$	M1M1 M1 A1 M1A1 [6]
		Total 7 marks

Type MS · What a log means: reading values and switching to index form

Answer 2 of 5 · Question 3 part a · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · What a log means: reading values and switching to index form

Answer 3 of 5 · Question 7 part a · 1 mark

2022 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

7(a)	4	B1 [1]
(b)	**Working in $\log_2$** $2\log_4 x = \frac{2\log_2 x}{\log_2 4} = \frac{2\log_2 x}{2} = [\log_2 x]$ $\log_2 16 + \frac{2\log_2 x}{2} = \log_2 y$ $\log_2 16x = \log_2 y$ OR $\log_2\left(\frac{x}{y}\right) = -4 \Rightarrow \frac{x}{y} = 2^{-4}$ $y = 16x^*$	M1 M1 M1 A1 cso [4]
	ALT	
	Working in $\log_4$ $\log_2 y = \frac{\log_4 y}{\log_4 2} = \frac{\log_4 y}{\frac{1}{2}} = 2\log_4 y = [\log_4 y^2]$ $\log_4 256 + \log_4 x^2 = \log_4 y^2$ $\log_4\left(256x^2\right) = \log_4 y^2$ OR $2\log_4\left(\frac{y}{x}\right) = 4 \Rightarrow \frac{y}{x} = 4^2$ $256x^2 = y^2 \Rightarrow y = 16x^*$	[M1 M1 M1 A1]
(c)	$16x = 4x + 5$ $16x = 4x + 5 \Rightarrow 12x = 5 \Rightarrow x = ...$ $x = \frac{5}{12}$	B1 M1 A1 [3]
		Total 8 marks

Type MS · What a log means: reading values and switching to index form

Answer 3 of 5 · Question 7 part a · 1 mark

2022 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Alt – working in $\log_4$		
	M1	For an attempt to change the base of $\log_2 y$ to base 4 using $\log_a y = \frac{\log_b y}{\log_b a}$ $\log_2 y = \frac{\log_4 y}{\log_4 2}\left[= \frac{\log_4 y}{\frac{1}{2}} = 2\log_4 y\right]$
	M1	For dealing with the indices **and** writing $4 = \log_4 256$ $\log_4 256 + \log_4 x^2 = \log_4 y^2$
	M1	Uses $\log A + \log B = \log AB$ to correctly combine the logs $\log_4\left(256x^2\right) = \log_4 y^2$ OR Uses $\log A - \log B = \log\frac{A}{B}$ to correctly combine the logs and removes logs $2\log_4\left(\frac{y}{x}\right) = 4$ **and** $\frac{y}{x} = 4^2$ (this approach will score the second and third M marks at this stage)
	A1	For correctly obtaining $y = 16x$*
(c)	**B1**	For writing down $16x = 4x + 5$
	M1	For an attempt to solve the equation $16x = 4x + 5 \Rightarrow 12x = 5 \Rightarrow x = ...$
	A1	For $x = \frac{5}{12}$
	Note: This is a 'hence' question. Condone candidates working without using given results, but the first mark is not awarded until the candidate reaches $16x = 4x + 5$	

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Answer 3 of 5 · Question 7 part a · 1 mark

2022 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · What a log means: reading values and switching to index form

Answer 4 of 5 · Question 10 part i · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part i in this mark scheme extract

10(i)	$\log_4(6y-5)=3 \Rightarrow 6y-5=64 \Rightarrow y=\frac{23}{2}$ oe	M1A1 [2]
(ii)	$\log_4(4-3x)^2-\frac{\log_4(x^2-5)}{\log_4 2}=3$ or $\frac{\log_2(4-3x)^2}{\log_2 4}-\log_2(x^2-5)=3$	M1
	$\frac{1}{2}\log_4(4-3x)^2-\log_4(x^2-5)=\frac{3}{2}$ or $\frac{1}{2}\log_2(4-3x)^2-\log_2(x^2-5)=3$	B1
	$\log_4(4-3x)-\log_4(x^2-5)=\frac{3}{2}$ or $\log_2(4-3x)-\log_2(x^2-5)=3$	M1
	$\log_4\left(\frac{4-3x}{x^2-5}\right)=\frac{3}{2}$ or $\log_2\left(\frac{4-3x}{x^2-5}\right)=3$	M1
	$\frac{4-3x}{x^2-5}=4^{\frac{3}{2}}=8$ or $\frac{4-3x}{x^2-5}=2^3=8$	M1
	$8x^2+3x-44=0$	A1
	$x=\frac{-3\pm\sqrt{3^2-4\times8\times-44}}{2\times8}=\frac{-3\pm\sqrt{1417}}{16} \Rightarrow \{x=2.1651...$ or $-2.540...\}$	M1A1
	$\therefore x=-2.54$ (awrt)	A1 [9]
		Total 11 marks

Type MS · What a log means: reading values and switching to index form

Answer 4 of 5 · Question 10 part i · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part i in this mark scheme extract

Part	Mark	Additional Guidance
(i)	M1	For a complete correct method to find the value of y
	A1	For the correct value
For part (ii) award the marks when you see them		
(ii)	M1	For change of base correctly to either 4 or 2
	B1	For replacing $\log_2 4$ with 2 or replacing $\log_4 2$ with $\frac{1}{2}$
	M1	For applying the power law correctly e.g.: $\frac{1}{2}\log_4(4-3x)^2 \Rightarrow \log_4(4-3x)$ or $\frac{1}{2}\log_2(4-3x)^2 \Rightarrow \log_2(4-3x)$
	M1	For applying the subtraction/addition law correctly
	M1	For correctly removed logs
	A1	For the correct 3TQ, "=0 "could be implied by later work
	M1	For a correct clear method to solve their 3TQ, allow one/two solutions from a calculator correct to 1dp (awrt) from a **correct** 3TQ
	A1	For two correct solutions, **exact** (allow embedded in a quadratic formula) or $x = 2.17$(awrt), $x = -2.54$(awrt) or JUST $x = -2.54$(awrt) seen only, from a correct 3TQ, scores M1A1A1(rejected 2.17 is implied)
	A1	For the correct solution. Accept awrt -2.54 The positive root must be rejected as it is not feasible.

Type MS · What a log means: reading values and switching to index form

Answer 4 of 5 · Question 10 part i · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part i in this mark scheme extract

Question number	Scheme	Marks

Type MS · What a log means: reading values and switching to index form

Answer 5 of 5 · Question 9 part a · 2 marks

2024 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

9 (a)	$\left(\log_a 8 = \frac{3}{4} \Rightarrow\right) a^{\frac{3}{4}} = 8 \left(\Rightarrow a = \left(\sqrt[3]{8}\right)^4\right) = 16$	M1A1 [2]
(b)	$\left(3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x =\right) 3x\log_2 x - \frac{4\log_2 8}{\log_2 16} + \frac{6x\log_2 8}{\log_2 4} - \log_2 x$	M1
	$\Rightarrow 3x\log_2 x - \log_2 8 + 3x\log_2 8 - \log_2 x$ $= (3x-1)\log_2 x + (3x-1)\log_2 8$ or $3x\log_2 8x - (1)\log_2 8x$	M1
	$\Rightarrow (3x-1)\log_2 8x$ or $\log_2 (8x)^{3x} + \log_2 (8x)^{-1} = \log_2 (8x)^{3x-1}$ *	M1A1 cso [4]
ALT	$\left(3x\log_2 x - 4\log_{16} 8 + 6x\log_4 8 - \log_2 x =\right) 3x\log_2 x - \frac{4\log_2 8}{\log_2 16} + \frac{6x\log_2 x}{\log_2 4} - \log_2 x$	M1
	$\Rightarrow 3x\log_2 x - \log_2 8 + 3x\log_2 8 - \log_2 x = \log_2 x^{3x} - \log_2 8 + \log_2 8^{3x} - \log_2 x$	M1
	$\left(\Rightarrow \log_2 (8x)^{3x} - \log_2 8x \Rightarrow\right) \log_2 \left(\frac{(8x)^{3x}}{8x}\right)$ or $\log_2 \left((8x)^{3x} \times (8x)^{-1}\right)$ or $\log_2 x^{3x-1} 8^{3x-1} = \log_2 (8x)^{3x-1}$ *	M1A1 cso [4]

"Box 3" of part b

We will see unanticipated methods once live marking begins.

If the answer is correct and **there is no incorrect working**, check the work carefully, to ascertain if they've shown enough steps to demonstrate use of the three main log laws this question tests and award full marks. If in any doubt at all – the response **MUST** be sent to review.

Other than this exception, please mark to the following rules.

Also use these rules if students don't gain the 2nd or 3rd M under the main or ALT schemes.

- M1 for any correct change of base to base 2
- M1 for any two correct applications of the power law or for
  $ax\log_2 8x + b\log_2 8x \Rightarrow \log_2 (8x)^{ax+b}$ or $(ax+b)\log_2 8x \Rightarrow \log_2 (8x)^{ax+b}$ $a, b \neq 0$
- M1 for any two correct applications of the addition or subtraction law

In each case – ignore any incorrect working.

Poor or incorrect bracketing may not be recovered in this question. (general principle of marking is usually that it can).

(c)	$\left[\log_2 (8x)^{3x-1} = 0 \Rightarrow \log_2 (8x)^{3x-1} = \log_2 8^0 \text{ or } (3x-1)\log_2 (8x) = 0\right]$	
	$\Rightarrow 3x - 1 = 0 \Rightarrow x = \frac{1}{3}$	M1A1
	$\Rightarrow 8x = 1 \Rightarrow x = \frac{1}{8}$	A1 [3]
		Total 9 marks

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Answer 5 of 5 · Question 9 part a · 2 marks

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Part	Mark	Notes
(a)	M1	For undoing the log to obtain $a^{\frac{3}{4}}=8$ or $(a=)8^{\frac{4}{3}}$
	A1	For $a = 16$ M1 A1 may be awarded for just seeing $a = 16$ unless from a string of incorrect work. Poor notation may be seen.
(b)	M1	For correctly changing the base to base 2 on at least one log, anywhere in their work. This may be implied and the mark awarded for eg $4\log_{16}8 \Rightarrow \frac{4\log_2 8}{\log_2 16}$ or $\frac{4\log_2 8}{4}$ or $\log_2 8$ or $\frac{12}{4}$ or 3 or $6x\log_4 8 \Rightarrow \frac{6x\log_2 8}{\log_2 4}$ or $\frac{6x\log_2 8}{2}$ or $3x\log_2 8$ or $9x$
	M1	For $(3x-1)\log_2 x+(3x-1)\log_2 8$
	M1	For correctly applying the power log law to their expression of the form $(ax+b)\log_2 8x \quad a,b\neq 0$ In general, poor bracketing may be recovered, but as this question is a show question, it generally cannot be recovered eg do not accept $8x^{3x}$
	A1 cso	For the correct expression, **minimum steps as shown** with no errors or omissions.
ALT	M1	For correctly changing the base to base 2 on at least one log. This may be implied and the mark awarded for $4\log_{16}8 \Rightarrow \frac{4\log_2 8}{\log_2 16}$ or $\frac{4\log_2 8}{4}$ or $\log_2 8$ or $\frac{12}{4}$ or 3 or $6x\log_4 8 \Rightarrow \frac{6x\log_2 8}{\log_2 4}$ or $\frac{6x\log_2 8}{2}$ or $3x\log_2 8$ or $9x$
	M1	For $\log_2 x^{(3x)}-\log_2 8+\log_2 8^{(3x)}-\log_2 x$
	M1	For correctly applying the subtraction law to an expression of the form (see the MS for minimum steps) $\log_2 x^{(ax)}-\log_2 8^b+\log_2 8^{(ax)}-\log_2 x^b$ The step in brackets doesn't need to be shown. Do not permit only eg $\log_2\left(x^{3x}\div 8\times 8^{3x}\div x\right)$ as a sufficient minimum step. In general, poor bracketing may be recovered, but as this question is a show question, it generally cannot be recovered eg do not accept $8x^{3x}\times 8x^{-1}$
	A1	For the correct expression, **minimum steps as shown** with no errors or omissions. .
	SC4	For working on both sides **with no errors** and achieving lhs = rhs Note, this is an exception to what we normally allow in a show that question (ie working on both sides until agreement)
(c)	M1	For setting $3x-1=0$ or $8x=1$ This mark may be implied by a correct answer.
	A1	For either $x=\frac{1}{3}$ or $\frac{1}{8}$
	A1	For both $x=\frac{1}{3}$ and $\frac{1}{8}$

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Answer 5 of 5 · Question 9 part a · 2 marks

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Question	Scheme	Marks